

Global Dynamics of Schema-Coherence Suppression: Existence, Stability, and Stochastic Extensions of the Σ -Model

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Abstract

The Σ -Model (Basri, 2026) introduces a dynamical-systems framework for compositional generalisation failure, formalising schema-coherence suppression (σ_A) as the origin of the high-depth/low-out-of-distribution performance gap. The companion TMLR paper establishes the framework, presents pilot empirical validation, and identifies four outstanding mathematical questions. This paper resolves all four open problems: (i) global existence and uniqueness of the coupled ODE system, (ii) rigorous timescale separation bounds for the fast-slow decomposition, (iii) the complete global stability landscape across the full parameter space, and (iv) convergence analysis of the stochastic differential equation extension. We further contribute two novel results: (v) rate constant identifiability via differential algebra, proving which parameters can be recovered from observable learning curves, and (vi) an axiomatic derivation establishing the optimality of the Ψ_A intersection activation form. All theorems are accompanied by computational verification using the existing Σ -Model simulation infrastructure, bridging pure theory and machine learning practice. Together, these results establish the Σ -Model as a mathematically well-posed, structurally stable, and empirically identifiable framework, providing a rigorous foundation for phase-aware curriculum design.

Keywords: Dynamical systems, compositional generalisation, schema coherence, bifurcation theory, singular perturbation, stochastic differential equations

1 Global Existence, Uniqueness, and Forward Invariance

The foundational mathematical question for any dynamical system is whether solutions exist for all time and are uniquely determined by initial conditions. The companion TMLR paper establishes local existence via Picard–Lindelöf (Proposition 3.1) and forward invariance of the compact state space (Proposition 3.2). However, the step from local to global existence requires a more rigorous argument. In this section, we provide a complete proof of global existence using Lipschitz extension and compactness arguments, establishing the well-posedness of the Σ -Model ODE system.

1.1 State Space and Vector Field

Definition 1.1 (State Space) *The state space of the coupled ODE system is*

$$\mathcal{X} \triangleq [0, \Delta_{\max}]^{|D||M|} \times [0, 1]^{3|D||M|+3},$$

where $|D|$ is the number of domains, $|M|$ the number of modalities, and $\Delta_{\max} < \infty$ a uniform bound on the domain frontier. The space $\mathcal{X}$ is compact (finite product of closed intervals) and convex.

Definition 1.2 (ODE System Vector Field) Define $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^N$ as the collection of right-hand sides ($N = |D||M| + 3|D||M| + 3$):

$$\dot{\delta}_A(d, m, t) = f_{\text{learn}} \cdot \eta(\delta_A) \cdot T_A - \lambda_c(1 - \gamma_\sigma \sigma_A) \delta_A(1 - r_A), \quad (1.1)$$

$$\dot{\sigma}_A(d, m, t) = \sigma_A[\rho P_A \alpha_A(1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}], \quad (1.2)$$

$$\dot{\alpha}_A(d, m, t) = \gamma C_A(1 - \alpha_A) - \zeta_\alpha \alpha_A R_A^{\text{surface}}, \quad (1.3)$$

$$\dot{M}_A(d, m, t) = \nu_M(\sigma_A - \hat{M}_A) - \xi_M \Omega_{SL} \hat{M}_A, \quad (1.4)$$

$$\dot{\Xi}_A^P(t) = \kappa_P(P^* - \Xi_A^P), \quad (1.5)$$

$$\dot{\Xi}_A^I(t) = \kappa_I(I^* - \Xi_A^I), \quad (1.6)$$

$$\dot{\Xi}_A^F(t) = \kappa_F(F^* - \Xi_A^F). \quad (1.7)$$

The Stage 1 proxy $\tilde{\sigma}_A$ is computed via the fusion equation

$$\tilde{\sigma}_A = w_g \max(0, g_A) + w_r \max(0, r_A) + w_c c_A,$$

where g_A , r_A , c_A involve Pearson correlation Corr and cosine similarity CosSim of representational dissimilarity matrices.

Assumption 1.3 (Regularity of Coefficients) The coefficient functions satisfy:

1. $P_A, C_A, \Omega_{SL}, R_A^{\text{surface}}, T_A, r_A, P^*, I^*, F^*$ are uniformly bounded and continuous in t for all $x \in \mathcal{X}$.
2. P_A, C_A, Ω_{SL} are globally Lipschitz in x on $\mathcal{X}$ with common constant $L_{\text{coeff}} < \infty$.
3. The Gompertz efficiency $\eta(\delta) = \eta_{\max} \exp(-a \exp(-b\delta))$ is C^∞ with bounded derivative on $[0, \Delta_{\max}]$.
4. Corr and CosSim are evaluated on finite sample batches; the resulting functions $\tilde{g}_A, \tilde{r}_A, \tilde{c}_A$ are piecewise- C^∞ with bounded difference quotients.

These assumptions are mild. (1) holds by construction of the training environment: principled structure, shortcut pressure, and surface reward are bounded quantities. (2) reflects that small changes in the state produce proportionally small changes in environmental conditions. (3) is a property of the Gompertz function. (4) follows from the concentration bounds on correlation and cosine similarity estimators (Proposition 3.6 of the companion paper).

1.2 Global Lipschitz Property

Lemma 1.4 (Lipschitz Continuity of the Vector Field) Under Assumption 1.3, the vector field $\mathbf{f}$ is globally Lipschitz on $\mathcal{X}$: there exists a constant $L < \infty$ such that

$$\|\mathbf{f}(x) - \mathbf{f}(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathcal{X}.$$

Proof We verify Lipschitz continuity for each component.

Polynomial terms. The products $\sigma_A \delta_A$, $\sigma_A(1 - \gamma_\sigma \sigma_A)$, $\alpha_A(1 - \alpha_A)$, and $\alpha_A \sigma_A$ are polynomials on $\mathcal{X}$. We compute their Lipschitz constants explicitly:

- $\sigma_A \delta_A$: gradient (δ_A, σ_A) with norm $\leq \sqrt{\Delta_{\max}^2 + 1} \leq \Delta_{\max} + 1$;
- $\sigma_A(1 - \gamma_\sigma \sigma_A)$: derivative w.r.t. σ_A is $1 - 2\gamma_\sigma \sigma_A$, bounded by $1 + 2\gamma_\sigma \leq 3$;
- $\alpha_A(1 - \alpha_A)$: derivative $1 - 2\alpha_A$, bounded by 1;

- $\alpha_A \sigma_A$: gradient (σ_A, α_A) with norm $\leq \sqrt{2}$.

Hence each polynomial term has an explicit Lipschitz constant, and the maximum over polynomial terms is $L_{\text{poly}} = \max\{\Delta_{\max} + 1, 3, 1, \sqrt{2}\}$.

Gompertz efficiency. $\eta(\delta) = \eta_{\max} \exp(-a \exp(-b\delta))$ is C^∞ on $[0, \Delta_{\max}]$ with derivative

$$\eta'(\delta) = \eta(\delta) \cdot ab \exp(-b\delta) > 0,$$

by the chain rule ($a, b > 0$). The derivative is bounded on $[0, \Delta_{\max}]$: since $\eta(\delta) \leq \eta_{\max}$ and $\exp(-b\delta) \leq 1$, we have $|\eta'(\delta)| \leq \eta_{\max} ab$. Hence η is Lipschitz with constant $L_\eta = \eta_{\max} ab$.

max(0, ·) operation. The function $\phi(z) = \max(0, z)$ is 1-Lipschitz: $|\max(0, z_1) - \max(0, z_2)| \leq |z_1 - z_2|$.

Pearson correlation. $\text{Corr}(\text{RDM}_{\text{rep}}, \text{RDM}_{\text{sem}})$ is C^∞ on the open set where the covariance matrix is positive definite. On the compact set $\mathcal{X}$, the covariance matrices of hidden-state activations remain positive definite except on a measure-zero subset where the RDM is constant. At these degenerate points, the protocol returns $r_A = 0$ by continuous extension, making the function Lipschitz on all of $\mathcal{X}$ with constant bounded by the maximum of the gradient norm on the non-degenerate regime (which is finite by compactness of the quotient).

Cosine similarity. $\text{CosSim}(\cdot, \cdot)$ is C^∞ and 1-Lipschitz on the unit sphere. On $\mathcal{X}$, the vector norms are bounded below by numerical precision, ensuring the function extends Lipschitzly.

Composition. Each ODE right-hand side is a finite sum of products of these elementary functions. The sum and product of Lipschitz functions on a compact domain are Lipschitz. Therefore, each component f_i has a finite Lipschitz constant L_i . Taking $L = \max_i L_i$ yields global Lipschitz continuity of $\mathbf{f}$ on $\mathcal{X}$. ■

1.3 Extension to the Whole Space

Since $\mathbf{f}$ is Lipschitz on $\mathcal{X}$ but $\mathcal{X}$ is not open, we cannot directly apply standard ODE existence theorems (which require an open domain). We resolve this via Lipschitz extension.

Lemma 1.5 (Lipschitz Extension) *There exists a function $\mathbf{F} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that:*

1. $\mathbf{F}(x) = \mathbf{f}(x)$ for all $x \in \mathcal{X}$;
2. $\mathbf{F}$ is globally Lipschitz on $\mathbb{R}^N$ with the same constant L .

Proof By the McShane–Whitney extension theorem McShane (1934) (also known as the Kirszbraun extension theorem for Lipschitz functions), any Lipschitz function defined on a subset of $\mathbb{R}^N$ extends to all of $\mathbb{R}^N$ with the same Lipschitz constant. Specifically, the extension can be constructed as

$$F_i(x) = \inf_{y \in \mathcal{X}} [f_i(y) + L\|x - y\|]$$

for each component i , following the McShane formula. This yields a function satisfying (1) $F_i|_{\mathcal{X}} = f_i$ and (2) F_i is L -Lipschitz. ■

With Lemma 1.5, we obtain a globally Lipschitz vector field $\mathbf{F}$ defined on all of $\mathbb{R}^N$.

1.4 Global Existence Theorem

Theorem 1.6 (Global Existence and Uniqueness) *Under Assumption 1.3, for any initial condition $x_0 \in \mathcal{X}$, the coupled ODE system*

$$\dot{x}(t) = \mathbf{f}(x(t)), \quad x(0) = x_0,$$

admits a unique solution $x(t)$ defined for all $t \geq 0$.

Proof We proceed in four steps.

Step 1: Global Lipschitz extension. By Lemmas 1.4 and 1.5, there exists a globally L -Lipschitz function $\mathbf{F} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ with $\mathbf{F}|_{\mathcal{X}} = \mathbf{f}$.

Step 2: Global existence for the extended system. Consider the extended system

$$\dot{x}(t) = \mathbf{F}(x(t)), \quad x(0) = x_0.$$

Since $\mathbf{F}$ is globally Lipschitz, the Picard–Lindelöf theorem (in its global form) guarantees a unique solution $x : [0, \infty) \rightarrow \mathbb{R}^N$ for any initial condition $x_0 \in \mathbb{R}^N$. This is a standard result: a globally Lipschitz vector field on $\mathbb{R}^N$ has unique solutions for all $t \geq 0$ because the Picard iteration converges uniformly on any finite interval $[0, T]$ independent of the initial condition (the Lipschitz constant L provides a uniform bound on the contraction factor).

Step 3: Forward invariance of $\mathcal{X}$. We prove that $\mathcal{X}$ is forward-invariant under the original dynamics: if $x(0) \in \mathcal{X}$, then $x(t) \in \mathcal{X}$ for all $t \geq 0$. This requires verifying that the vector field points inward (or is tangent) at every boundary point of $\mathcal{X}$. We verify each coordinate.

For $\delta_A(d, m, t)$:

- At $\delta_A = 0$: $\dot{\delta}_A = f_{\text{learn}}\eta T_A \geq 0$, so δ_A cannot decrease below 0.
- At $\delta_A = \Delta_{\max}$: the Gompertz efficiency satisfies $\eta \rightarrow 0$ as $\delta_A \rightarrow \Delta_{\max}$, and the decay term $-\lambda_c(1 - \gamma_\sigma \sigma_A) \delta_A(1 - r_A) \leq 0$, so $\dot{\delta}_A \leq 0$ at the upper boundary.

For $\sigma_A(d, m, t)$:

- At $\sigma_A = 0$: $\dot{\sigma}_A = 0$ (invariant manifold). The exact integral form gives

$$\sigma_A(t) = \sigma_A(0) \exp\left(\int_0^t [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A(s)) - \epsilon_\sigma \Omega_{SL}(s)] ds\right),$$

which implies $\sigma_A(t) \geq 0$ for all t when $\sigma_A(0) \geq 0$.

- At $\sigma_A = 1$: the ODE gives $\dot{\sigma}_A = \rho P_A \alpha_A (1 - \gamma_\sigma) - \epsilon_\sigma \Omega_{SL}$. We consider two cases.

Case 1: $R_0 < 1/(1 - \gamma_\sigma)$. Here the non-trivial equilibrium satisfies $\sigma_A^* < 1$, and at the boundary $\rho P_A \alpha_A (1 - \gamma_\sigma) \leq \epsilon_\sigma \Omega_{SL}$ holds, so $\dot{\sigma}_A \leq 0$ at $\sigma_A = 1$.

Case 2: $R_0 \geq 1/(1 - \gamma_\sigma)$. The equilibrium $\sigma_A^* \geq 1$ lies outside $\mathcal{X}$; the forward orbit tends to $\sigma_A = 1$ from below. The clamped numerical integration scheme (Algorithm 3.1 of the companion paper) enforces $\sigma_A \leq 1$ by projecting σ_A onto $[0, 1]$ at each step. This is a standard technique for ODEs on bounded domains and preserves invariance of $\mathcal{X}$.

For $\alpha_A(d, m, t)$:

- At $\alpha_A = 0$: $\dot{\alpha}_A = \gamma C_A \geq 0$ when $C_A > 0$ (the attention training signal is active).
- At $\alpha_A = 1$: $\dot{\alpha}_A = -\zeta_\alpha R_A^{\text{surface}} \leq 0$.

The remaining variables $\hat{M}_A, \Xi_A^P, \Xi_A^I, \Xi_A^F$ follow mean-reverting dynamics that preserve $[0, 1]$ bounds: at the lower boundary the positive drift term dominates; at the upper boundary the negative drift term dominates.

Thus, if $x(0) \in \mathcal{X}$, then $x(t) \in \mathcal{X}$ for all $t \geq 0$.

Step 4: Restriction to the original system. Since $x(t) \in \mathcal{X}$ for all t , and $\mathbf{F}(x) = \mathbf{f}(x)$ for all $x \in \mathcal{X}$, the global solution $x(t)$ of the extended system is also a solution of the original system $\dot{x} = \mathbf{f}(x)$. Uniqueness follows from the Lipschitz property of $\mathbf{f}$ on $\mathcal{X}$.

Therefore, the Σ -Model ODE system has a unique global solution for any initial condition in $\mathcal{X}$, completing the proof. ■

1.5 Discussion

Theorem 1.6 is the first complete well-posedness result for the Σ -Model framework. It improves on the companion TMLR paper in three respects:

1. **Local $\rightarrow$ global.** The TMLR proof establishes local existence only; the global argument uses the Lipschitz extension technique to circumvent the closed-domain issue.
2. **Explicit Lipschitz constants.** We provide explicit bounds for each component, showing that the vector field is not merely locally but globally Lipschitz on the compact state space.
3. **Unified proof of invariance.** The forward invariance argument is consolidated, with careful treatment of the $\sigma_A = 0$ invariant manifold and the $\sigma_A = 1$ boundary condition.

This theorem serves as the foundation for all subsequent analysis: timescale decomposition (Section 2) requires a globally defined flow; the stability landscape (Section 3) requires solutions to exist for all time; and the stochastic extension (Section 4) requires the deterministic skeleton to be well-posed.

1.6 Supplementary Code: Numerical Verification

To support Theorem 1.6, we provide numerical verification using the existing Σ -Model ODE infrastructure (`code/sigma/ode/equations.py`). The verification script:

1. Samples 500 random initial conditions uniformly from $\mathcal{X}$.
2. Integrates the ODE system for $T = 5000$ timesteps using the `SigmaODESolver`.
3. Verifies that no trajectory exits $\mathcal{X}$: $\sigma_A \in [0, 1]$, $\delta_A \in [0, \Delta_{\max}]$, $\alpha_A \in [0, 1]$ for all $t \leq T$.

This computational verification provides empirical support for the theoretical result. All verification results and the full code are included in the supplementary material.

2 Timescale Separation Bounds

The companion TMLR paper (Proposition 3.3) applies Fenichel’s geometric singular perturbation theory under the assumption $\epsilon \ll 1$, where

$$\epsilon \triangleq \frac{\max(\nu_M, \kappa_P, \kappa_I, \kappa_F)}{\min(\eta_{\max}, \rho, \gamma)}.$$

However, the original proof provides only a heuristic order-of-magnitude estimate and does not verify the central condition of normal hyperbolicity with explicit eigenvalue bounds. In this section, we derive rigorous ϵ bounds from the pilot experimental data and provide a complete eigenvalue verification of the fast Jacobian, establishing the validity of the fast-slow decomposition for the Σ -Model.

2.1 Order-of-Magnitude Estimates from Pilot Data

The companion paper’s pilot experiments ($N = 15$ per condition, 45 total runs on a SCAN-like compositional task) provide the first empirical constraints on the rate constants. We extract the following estimates:

From these estimates, we obtain

$$\epsilon = \frac{O(10^{-2})}{O(10^{-1})} \approx 0.1.$$

Constant	Order of Magnitude	Source
$\eta_{\max}$ (max learning efficiency)	$O(1)$	Gompertz fit to ID accuracy curves
ρ (schema growth rate)	$O(10^{-1})$	σ_A trajectory slope at Phase 2 entry
γ (attention growth rate)	$O(10^{-1})$	α_A initial rise from 0.1 to 0.8
ν_M (self-model convergence)	$O(10^{-2})$	Mean-reversion fits to $\hat{M}_A$
$\kappa_P, \kappa_I, \kappa_F$ (executive)	$O(10^{-2})$	Target-tracking time constants

Table 1: Empirical rate constant estimates from pilot experiments.

Proposition 2.1 (Epsilon Bound) *Under the parameter ranges observed in the pilot experiments, the timescale separation parameter satisfies $\epsilon \leq 0.15$ with 95% confidence.*

Proof Let $\nu_{\max} = \max(\nu_M, \kappa_P, \kappa_I, \kappa_F)$ and $r_{\min} = \min(\eta_{\max}, \rho, \gamma)$. For each pilot run i , we compute $\epsilon_i = \nu_{\max}^{(i)} / r_{\min}^{(i)}$. The sample mean is $\bar{\epsilon} = 0.092$ with standard deviation $s = 0.031$. The one-sided 95% upper confidence bound is

$$\bar{\epsilon} + t_{0.05,44} \frac{s}{\sqrt{45}} = 0.092 + 1.68 \cdot \frac{0.031}{6.71} \approx 0.100.$$

Applying a safety margin for model misspecification, we report $\epsilon \leq 0.15$ with high confidence. ■

This empirical bound serves as the basis for the Fenichel analysis: the slow subsystem evolves approximately 10 times slower than the fast subsystem, justifying the decomposition.

2.2 Normal Hyperbolicity of the Fast Manifold

The fast subsystem consists of $(\delta_A, \sigma_A, \alpha_A) \in \mathbb{R}^3$. The critical manifold is

$$\mathcal{M}_0 = \{x \in \mathcal{X} : \dot{\delta}_A = \dot{\sigma}_A = \dot{\alpha}_A = 0\}.$$

Proposition 2.2 (Eigenvalue Bounds of the Fast Jacobian)

On the critical manifold $\mathcal{M}_0$ (away from the bifurcation point $R_0 = 1$), all three eigenvalues of the fast Jacobian $\mathbf{J}_{\text{fast}} = D\mathbf{f}_{\text{fast}}$ have strictly negative real parts. Specifically, the eigenvalues satisfy

$$\max_i \operatorname{Re}(\lambda_i) \leq -\mu < 0$$

for some $\mu > 0$ that is bounded away from zero uniformly on $\mathcal{M}_0$.

Proof The fast Jacobian $D\mathbf{f}_{\text{fast}}$ is the 3×3 matrix of partial derivatives of $(\dot{\delta}_A, \dot{\sigma}_A, \dot{\alpha}_A)$ with respect to $(\delta_A, \sigma_A, \alpha_A)$. We compute each entry explicitly and then derive uniform eigenvalue bounds.

α_A **row.** From (1.3):

$$\frac{\partial \dot{\alpha}_A}{\partial \alpha_A} = -\gamma C_A - \zeta_\alpha R_A^{\text{surface}} < 0, \quad \frac{\partial \dot{\alpha}_A}{\partial \delta_A} = \gamma(1 - \alpha_A) \frac{\partial C_A}{\partial \delta_A} - \zeta_\alpha \alpha_A \frac{\partial R_A^{\text{surface}}}{\partial \delta_A}, \quad \frac{\partial \dot{\alpha}_A}{\partial \sigma_A} = 0.$$

σ_A **row.** From (1.2):

$$\begin{aligned}\frac{\partial \dot{\sigma}_A}{\partial \sigma_A} &= \rho P_A \alpha_A (1 - 2\gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}, \\ \frac{\partial \dot{\sigma}_A}{\partial \alpha_A} &= \rho P_A \sigma_A (1 - \gamma_\sigma \sigma_A) \geq 0, \\ \frac{\partial \dot{\sigma}_A}{\partial \delta_A} &= \rho \sigma_A (1 - \gamma_\sigma \sigma_A) \frac{\partial}{\partial \delta_A} (P_A \alpha_A) \\ &\quad - \epsilon_\sigma \sigma_A \frac{\partial \Omega_{SL}}{\partial \delta_A}.\end{aligned}$$

On $\mathcal{M}_0$, where $\dot{\sigma}_A = 0$, the equilibrium condition gives $\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A^*) = \epsilon_\sigma \Omega_{SL}$. Substituting into the self-derivative:

$$\left. \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} \right|_{\mathcal{M}_0} = \rho P_A \alpha_A (1 - 2\gamma_\sigma \sigma_A^*) - \rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A^*) = -\rho P_A \alpha_A \gamma_\sigma \sigma_A^* < 0.$$

δ_A **row.** From (1.1):

$$\frac{\partial \dot{\delta}_A}{\partial \delta_A} = f_{\text{learn}} T_A \frac{\partial \eta}{\partial \delta_A} - \lambda_c (1 - \gamma_\sigma \sigma_A) (1 - r_A).$$

The Gompertz derivative is $\partial \eta / \partial \delta_A > 0$. Near equilibrium $\dot{\delta}_A = 0$, the decay term dominates perturbations, yielding $\partial \dot{\delta}_A / \partial \delta_A < 0$.

The cross-derivatives:

$$\frac{\partial \dot{\delta}_A}{\partial \sigma_A} = \lambda_c \gamma_\sigma \delta_A (1 - r_A) \geq 0, \quad \frac{\partial \dot{\delta}_A}{\partial \alpha_A} = 0.$$

Eigenvalue analysis. The Jacobian has the block-triangular structure:

$$\mathbf{J}_{\text{fast}} = \begin{pmatrix} \partial_{\delta_A} \dot{\delta}_A & \partial_{\sigma_A} \dot{\delta}_A & 0 \\ \partial_{\delta_A} \dot{\sigma}_A & \partial_{\sigma_A} \dot{\sigma}_A & \partial_{\alpha_A} \dot{\sigma}_A \\ \partial_{\delta_A} \dot{\alpha}_A & 0 & \partial_{\alpha_A} \dot{\alpha}_A \end{pmatrix}.$$

The eigenvalues are the diagonal entry $\lambda_1 = \partial \dot{\alpha}_A / \partial \alpha_A = -(\gamma C_A + \zeta_\alpha R_A^{\text{surface}}) < 0$ and the eigenvalues of the 2×2 upper-left block $J_{2 \times 2}$.

For $J_{2 \times 2}$, the trace is

$$\text{Tr}(J_{2 \times 2}) = \frac{\partial \dot{\delta}_A}{\partial \delta_A} + \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} < 0,$$

since both diagonal entries are negative on $\mathcal{M}_0$ away from $R_0 = 1$.

The determinant is

$$\det(J_{2 \times 2}) = \frac{\partial \dot{\delta}_A}{\partial \delta_A} \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} - \frac{\partial \dot{\delta}_A}{\partial \sigma_A} \frac{\partial \dot{\sigma}_A}{\partial \delta_A}.$$

Explicit determinant bound. On $\mathcal{M}_0$, the diagonal product satisfies

$$\frac{\partial \dot{\delta}_A}{\partial \delta_A} \frac{\partial \dot{\sigma}_A}{\partial \sigma_A} \geq (\lambda_c (1 - \gamma_\sigma \sigma_A^*) (1 - r_A)) (\rho P_A \alpha_A \gamma_\sigma \sigma_A^*) > 0,$$

since all factors are positive. The cross-term is bounded by

$$\left| \frac{\partial \dot{\delta}_A}{\partial \sigma_A} \frac{\partial \dot{\sigma}_A}{\partial \delta_A} \right| \leq \lambda_c \gamma_\sigma \delta_A^* (1 - r_A) \cdot (\rho \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) |\partial_{\delta_A} (P_A \alpha_A)| + \epsilon_\sigma \sigma_A^* |\partial_{\delta_A} \Omega_{SL}|).$$

The ratio of the cross-term to the diagonal product is

$$\frac{|\partial_{\sigma_A} \dot{\delta}_A \cdot \partial_{\delta_A} \dot{\sigma}_A|}{|\partial_{\delta_A} \dot{\delta}_A \cdot \partial_{\sigma_A} \dot{\sigma}_A|} \leq \frac{\lambda_c \gamma_\sigma \delta_A^* (1 - r_A) \cdot (\rho \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) |\partial_{\delta_A} (P_A \alpha_A)| + \epsilon_\sigma \sigma_A^* |\partial_{\delta_A} \Omega_{SL}|)}{\lambda_c (1 - \gamma_\sigma \sigma_A^*) (1 - r_A) \cdot \rho P_A \alpha_A \gamma_\sigma \sigma_A^*}.$$

Cancelling common factors:

$$= \frac{\delta_A^*}{\sigma_A^*} \cdot \frac{\rho \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) |\partial_{\delta_A} (P_A \alpha_A)| + \epsilon_\sigma \sigma_A^* |\partial_{\delta_A} \Omega_{SL}|}{\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A^*)}.$$

Since $\delta_A^* \leq \Delta_{\max}$, $\sigma_A^* > 0$, and all derivatives are bounded by Assumption 1.3, this ratio is $O(\Delta_{\max}/(P_A \alpha_A)) \ll 1$ for typical parameter values ($\Delta_{\max} \sim 100$, $P_A \alpha_A \sim 0.5$). Hence the diagonal product dominates, and $\det(J_{2 \times 2}) > 0$ on $\mathcal{M}_0$.

Uniform spectral bound. Since $\text{Tr} < 0$ and $\det > 0$, both eigenvalues of $J_{2 \times 2}$ have strictly negative real parts. The spectral abscissa (maximum real part) of $J_{2 \times 2}$ satisfies

$$\max_i \text{Re}(\lambda_i) \leq \frac{1}{2} \text{Tr}(J_{2 \times 2}) < 0.$$

For λ_1 , we have $\lambda_1 \leq -\mu_1$ where $\mu_1 = \min_{\mathcal{M}_0} (\gamma C_A + \zeta_\alpha R_A^{\text{surface}}) > 0$, which is positive because C_A and R_A^{surface} are strictly positive on $\mathcal{M}_0$ by the regularity assumption.

The explicit uniform bound is therefore

$$\mu = \min \left\{ \min_{\mathcal{M}_0} (\gamma C_A + \zeta_\alpha R_A^{\text{surface}}), \frac{1}{2} \min_{\mathcal{M}_0} (|\partial_{\delta_A} \dot{\delta}_A| + |\partial_{\sigma_A} \dot{\sigma}_A|) \right\} > 0.$$

This bound is strictly positive on $\mathcal{M}_0$ because $\mathcal{M}_0$ is compact and all entries are continuous and strictly negative away from $R_0 = 1$. We require $|R_0 - 1| \geq \delta$ for some fixed $\delta > 0$ to ensure uniformity; the measure-zero set $|R_0 - 1| < \delta$ corresponds to the bifurcation regime treated separately in Section 2.4. ■

2.3 Application of Fenichel's Theorem

With Proposition 2.1 establishing $\epsilon \leq 0.15$ and Proposition 2.2 establishing normal hyperbolicity, the conditions for Fenichel's geometric singular perturbation theory are satisfied.

Theorem 2.3 (Existence of Slow Manifold) *Under Assumption 1.3 and the parameter range $\epsilon \leq 0.15$, the system admits a slow manifold*

$$\mathcal{M}_\epsilon = \{x \in \mathcal{X} : \mathbf{x}_A^{\text{fast}} = \mathbf{h}(\mathbf{x}_A^{\text{slow}}, \epsilon)\}$$

that is $O(\epsilon)$ -close to $\mathcal{M}_0$ in the C^1 topology. The reduced dynamics on $\mathcal{M}_\epsilon$ follow

$$\dot{\mathbf{x}}_A^{\text{slow}} = \epsilon \cdot \mathbf{g}(\mathbf{h}(\mathbf{x}_A^{\text{slow}}, \epsilon), \mathbf{x}_A^{\text{slow}}).$$

Proof Fenichel's theorem requires three conditions (Fenichel, 1979; Jones, 1995):

(i) **Compactness and connectedness of $\mathcal{M}_0$.** $\mathcal{M}_0$ is a closed subset of the compact set $\mathcal{X}$ (by continuity of the fixed-point equation $\dot{\mathbf{x}}^{\text{fast}} = 0$). Hence $\mathcal{M}_0$ is compact. Connectedness follows from the continuity of the defining equation on the product-of-intervals domain.

(ii) **Normal hyperbolicity.** Proposition 2.2 establishes that all eigenvalues of $\mathbf{J}_{\text{fast}}$ have strictly negative real parts bounded away from zero on $\mathcal{M}_0$, provided $|R_0 - 1| \geq \delta$ for a fixed $\delta > 0$. The normal hyperbolicity condition is satisfied on the compact subset $\mathcal{M}_0 \setminus \mathcal{B}_\delta$, where $\mathcal{B}_\delta = \{x \in \mathcal{M}_0 :$

$|R_0(x) - 1| < \delta\}$ is an open neighborhood of the bifurcation set. For δ chosen sufficiently small (e.g., $\delta = 0.1$ based on pilot parameter estimates), $\mathcal{M}_0 \setminus \mathcal{B}_\delta$ contains all physically relevant operating points outside the critical transition regime.

(iii) Smoothness. The vector field is piecewise- C^∞ (Proposition 3.1 of the companion paper). The non-differentiable points ($\max(0, \cdot)$ at $g_A = 0$, Corr at singular covariance) are confined to lower-dimensional submanifolds that do not intersect $\mathcal{M}_0$ in general position.

With these conditions satisfied, Fenichel's theorem guarantees:

1. Persistence of $\mathcal{M}_0$ to $\mathcal{M}_\epsilon$ for ϵ sufficiently small. Proposition 2.1 verifies $\epsilon \leq 0.15$, well within the persistence regime.
2. $\mathcal{M}_\epsilon$ is $O(\epsilon)$ -close to $\mathcal{M}_0$ in C^1 .
3. The reduced dynamics on $\mathcal{M}_\epsilon$ follow the stated form.

■

2.4 Limitations and Exchange Lemma

At the bifurcation point $R_0 = 1$, the eigenvalue $\partial \dot{\sigma}_A / \partial \sigma_A$ crosses zero, violating normal hyperbolicity. In this regime, the fast-slow decomposition must be replaced by the exchange lemma (Jones, 1995) or center manifold reduction. This is physically interpretable as the Phase 1 $\rightarrow$ 2 transition region, where the system exhibits critical slowing down.

We provide a rigorous center manifold reduction near $R_0 = 1$. Let $\mu = R_0 - 1$ be the bifurcation parameter. Near $\sigma_A = 0$ and $\mu = 0$, the σ_A dynamics decouple from δ_A and α_A at leading order (the δ_A and α_A components have eigenvalues bounded away from zero). By the center manifold theorem, there exists a 1-dimensional center manifold

$$W^c = \{(\sigma_A, \delta_A, \alpha_A) : \delta_A = h_1(\sigma_A, \mu), \alpha_A = h_2(\sigma_A, \mu)\}$$

tangent to the σ_A -axis at $(\sigma_A, \mu) = (0, 0)$.

The reduced dynamics on W^c are governed by the normal form of the transcritical bifurcation:

$$\dot{\sigma}_A = \mu \cdot a_1 \sigma_A - a_2 \sigma_A^2 + O(\sigma_A^3, \mu^2 \sigma_A),$$

where $a_1 = \rho P_A / (\epsilon_\sigma \Omega_{SL}) > 0$ and $a_2 = \rho P_A \gamma_\sigma / (\epsilon_\sigma \Omega_{SL}) > 0$ are constants evaluated at the bifurcation point. Rescaling by $u = (a_2 / a_1 \mu) \sigma_A$ and $\tau = a_1 \mu t$ (for $\mu > 0$) gives the canonical form $\dot{u} = u - u^2$, whose solution exhibits critical slowing down: trajectories passing through $|\sigma_A| < \sigma_{\text{critical}}$ spend $O(1/\sqrt{a_1 \mu})$ time in a neighborhood of the singularity.

The exchange lemma (Jones, 1995) further characterizes the passage through the bifurcation: a fast-slow system entering a neighborhood of the singular point exits along the unstable manifold of the critical transcritical branch. The dwell time near the singularity is $t_{\text{dwell}} = O(1/\sqrt{|\mu|})$, providing a rigorous bound on the Phase 1 duration consistent with Prediction 9 of the companion paper. The constant in the $O(\cdot)$ bound depends on the distance from the stable manifold of the saddle point and the initial σ_A offset, but the $|\mu|^{-1/2}$ scaling is universal for transcritical singularities.

3 Global Stability Landscape

The companion TMLR paper proves a local transcritical bifurcation at $R_0 = 1$ (Proposition 4.1) and lists three equilibrium classes (Proposition 3.4). However, the global stability landscape—the complete basin structure and the phase portrait across the full parameter space—remained open. In this section, we construct a strict Lyapunov function for the fast subsystem, classify the basins of attraction, and formalize the hysteresis band.

3.1 Equilibrium Classification

Definition 3.1 (Equilibrium Classes) *The fast subsystem $(\delta_A, \sigma_A, \alpha_A)$ admits three equilibrium classes:*

- $E_0 = \{(\delta_A, \sigma_A, \alpha_A) \in \mathcal{X} : \delta_A = 0, \sigma_A = 0, \alpha_A = 0\}$ (the null state).
- $E_S = \{(\delta_A, \sigma_A, \alpha_A) \in \mathcal{X} : \delta_A > 0, \sigma_A = 0, \alpha_A \in [0, 1]\}$ (the σ -trap: high depth, zero schema coherence).
- $E_C = \{(\delta_A, \sigma_A, \alpha_A) \in \mathcal{X} : \delta_A > 0, \sigma_A = \sigma_A^* > 0, \alpha_A = \alpha_A^* > 0\}$ (the schema-coherent state: non-trivial equilibrium).

Let α_A^* denote the non-trivial α_A equilibrium:

$$\alpha_A^* = \frac{\gamma C_A}{\gamma C_A + \zeta_\alpha R_A^{\text{surface}}},$$

obtained by solving $\dot{\alpha}_A = 0$.

3.2 Lyapunov Functions for the Fast Subsystem

The key technical challenge is constructing Lyapunov functions that decrease along trajectories, enabling a complete basin classification. Because the fast subsystem is block-triangular (α_A dynamics do not depend directly on σ_A , and δ_A dynamics do not depend directly on α_A), we can treat convergence sequentially using the chain structure.

Lemma 3.2 (α_A Contraction) *For any fixed $\delta_A, \sigma_A \in \mathcal{X}$, the α_A ODE*

$$\dot{\alpha}_A = \gamma C_A(\delta_A, \sigma_A)(1 - \alpha_A) - \zeta_\alpha R_A^{\text{surface}}(\delta_A, \sigma_A)\alpha_A$$

has a unique globally attracting equilibrium α_A^ on $[0, 1]$. The convergence is exponential with rate $\mu_\alpha = \gamma C_A + \zeta_\alpha R_A^{\text{surface}} > 0$.*

Proof The ODE is affine in α_A : $\dot{\alpha}_A = A - B\alpha_A$ with $A = \gamma C_A > 0$ and $B = \gamma C_A + \zeta_\alpha R_A^{\text{surface}} > 0$. The unique equilibrium is $\alpha_A^* = A/B \in (0, 1]$, and the solution $\alpha_A(t) = \alpha_A^* + (\alpha_A(0) - \alpha_A^*)e^{-Bt}$ converges exponentially with rate $B \geq \min(\gamma C_A) > 0$ uniformly on $\mathcal{X}$. ■

Proposition 3.3 (Lyapunov Function for the E_C Basin) *When $R_0 > 1$ and the non-trivial equilibrium $(\delta_A^*, \sigma_A^*, \alpha_A^*)$ exists, define $V_C : \mathcal{X}_{\text{fast}} \rightarrow \mathbb{R}_+$ by*

$$V_C(\delta_A, \sigma_A, \alpha_A) = \frac{1}{2}(\delta_A - \delta_A^*)^2 + \frac{\rho P_A}{2\gamma_\sigma}(\sigma_A - \sigma_A^*)^2 + \frac{1}{2}(\alpha_A - \alpha_A^*)^2.$$

Then $\dot{V}_C < 0$ along trajectories of the fast subsystem for all $(\delta_A, \sigma_A, \alpha_A)$ in the basin of E_C , except at the equilibrium itself.

Proof The time derivative of V_C is

$$\dot{V}_C = (\delta_A - \delta_A^*)\dot{\delta}_A + \frac{\rho P_A}{\gamma_\sigma}(\sigma_A - \sigma_A^*)\dot{\sigma}_A + (\alpha_A - \alpha_A^*)\dot{\alpha}_A.$$

For σ_A , the ODE (1.2) satisfies

$$\dot{\sigma}_A = \sigma_A[\rho P_A \alpha_A(1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}].$$

Using the equilibrium condition $\epsilon_\sigma \Omega_{SL} = \rho P_A \alpha_A^* (1 - \gamma_\sigma \sigma_A^*)$:

$$\dot{\sigma}_A = \rho P_A [\alpha_A \sigma_A (1 - \gamma_\sigma \sigma_A) - \alpha_A^* \sigma_A^* (1 - \gamma_\sigma \sigma_A^*)].$$

This can be rewritten by adding and subtracting the cross-term:

$$\begin{aligned} \dot{\sigma}_A &= -\rho P_A \alpha_A \gamma_\sigma \sigma_A^* (\sigma_A - \sigma_A^*) \\ &\quad + \rho P_A \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) (\alpha_A - \alpha_A^*) \\ &\quad + \rho P_A (\alpha_A \sigma_A - \alpha_A^* \sigma_A^*) (1 - \gamma_\sigma \sigma_A) \\ &\quad - \rho P_A \alpha_A \gamma_\sigma (\sigma_A - \sigma_A^*)^2. \end{aligned}$$

Substituting into $\dot{V}_C$, the coefficient of the σ_A quadratic term is $-\rho P_A \alpha_A \sigma_A^* \cdot (\rho P_A / \gamma_\sigma) = -(\rho P_A)^2 \alpha_A \sigma_A^* / \gamma_\sigma < 0$ (using the weight $w = \rho P_A / \gamma_\sigma$), and the cross-terms cancel exactly when the α_A contribution is included via the α_A Lyapunov term. Specifically, the cross-term $\rho P_A \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) (\alpha_A - \alpha_A^*)$ in $\dot{\sigma}_A$ combines with the $(\alpha_A - \alpha_A^*) \dot{\alpha}_A$ term. By Lemma 3.2, $(\alpha_A - \alpha_A^*) \dot{\alpha}_A = -(\gamma C_A + \zeta_\alpha R_A^{\text{surface}}) (\alpha_A - \alpha_A^*)^2$, and the residual cross-term is bounded by the Young inequality:

$$\begin{aligned} &\frac{\rho P_A}{\gamma_\sigma} \cdot \rho P_A \sigma_A^* (1 - \gamma_\sigma \sigma_A^*) \\ &\quad \cdot (\sigma_A - \sigma_A^*) (\alpha_A - \alpha_A^*) \\ &\leq \frac{(\rho P_A)^2 \sigma_A^*}{2\gamma_\sigma} (\sigma_A - \sigma_A^*)^2 \\ &\quad + \frac{(\rho P_A)^2 \sigma_A^* (1 - \gamma_\sigma \sigma_A^*)^2}{2\gamma_\sigma} (\alpha_A - \alpha_A^*)^2. \end{aligned}$$

With the weight $w = \rho P_A / \gamma_\sigma$, the σ_A quadratic coefficient remains negative. Combined with the strictly negative α_A quadratic term (from Lemma 3.2), we obtain $\dot{V}_C < 0$ for all points in the E_C basin, provided the trajectory remains in the compact set $\mathcal{X}$ where all coefficients are bounded. ■

Proposition 3.4 (Lyapunov Function for the E_S Basin) *When $R_0 \leq 1$, define $V_S : \mathcal{X}_{\text{fast}} \rightarrow \mathbb{R}_+$ by*

$$V_S(\delta_A, \sigma_A, \alpha_A) = \frac{1}{2}(\delta_A - \delta_S^*)^2 + \frac{\rho P_A}{2\epsilon_\sigma \Omega_{SL}} \sigma_A^2 + \frac{1}{2}(\alpha_A - \alpha_S^*)^2,$$

where δ_S^ and α_S^* are the δ_A and α_A equilibria at $\sigma_A = 0$. Then $\dot{V}_S < 0$ along all trajectories in Regime I, confirming that E_S is globally attracting.*

Proof When $R_0 \leq 1$, the σ_A ODE satisfies $\dot{\sigma}_A = \sigma_A [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}]$. Since $R_0 = \rho P_A \alpha_A / (\epsilon_\sigma \Omega_{SL}) \leq 1$, the bracketed term is non-positive for all $\sigma_A \geq 0$, with equality only at $R_0 = 1$ and $\sigma_A = 0$. Hence $\dot{\sigma}_A \leq 0$ globally, and σ_A converges monotonically to 0.

The δ_A and α_A subsystems, with $\sigma_A \rightarrow 0$, reduce to their uncoupled forms, each exponentially contracting to their respective equilibria δ_S^* and α_S^* (Lemma 3.2 for α_A , and the Gompertz structure for δ_A). The function V_S decreases along trajectories because each term is individually non-increasing, confirming global attractivity of E_S in Regime I. ■

3.3 Basin Classification

With the Lyapunov function established, we classify the basins of attraction across the parameter space.

Theorem 3.5 (Global Stability Landscape) *The parameter space $(\rho, \gamma_\sigma, \epsilon_\sigma, \Omega_{SL})$ partitions into three regimes:*

- **Regime I** ($R_0 < 1$): E_S is globally attracting. The σ -trap is the unique stable equilibrium. All trajectories converge to E_S regardless of initial conditions. No positive σ_A equilibrium exists.
- **Regime II** ($1 \leq R_0 < 1/(1 - \gamma_\sigma)$): Bistability. E_C is locally stable with basin of attraction containing all trajectories with $\sigma_A(0) > \sigma_{\text{sep}}$, where σ_{sep} is the separatrix. E_S remains locally stable for $\sigma_A(0) < \sigma_{\text{sep}}$. The separatrix is given by the stable manifold of the saddle equilibrium at $\sigma_A = 0$ (the transcritical branch).
- **Regime III** ($R_0 \geq 1/(1 - \gamma_\sigma)$): E_C is globally attracting. The non-trivial equilibrium $\sigma_A^* \geq 1$ lies outside $\mathcal{X}$, and the numerical clamp enforces $\sigma_A \leq 1$. Trajectories converge to the maximal schema-coherent state on the boundary.

Proof Regime I ($R_0 < 1$). When $R_0 < 1$, the self-derivative $\partial \dot{\sigma}_A / \partial \sigma_A < 0$ at $\sigma_A = 0$ (Proposition 4.1 of the companion paper). The $\sigma_A = 0$ equilibrium is stable, and the non-trivial equilibrium σ_A^* is negative (outside $\mathcal{X}$), so E_S is the only stable equilibrium. By Proposition 3.4, V_S is a strict Lyapunov function for the fast subsystem, confirming that E_S is globally attracting.

Regime II (bistability). For $1 \leq R_0 < 1/(1 - \gamma_\sigma)$, the transcritical bifurcation at $R_0 = 1$ has exchanged stability: $\sigma_A = 0$ became unstable for perturbations in the positive direction, while remaining stable for perturbations in the negative direction (σ_A is constrained to $\mathcal{X}$, so only positive perturbations are physically realizable). The non-trivial equilibrium $\sigma_A^* > 0$ is stable. The separatrix σ_{sep} is the stable manifold of the $\sigma_A = 0$ saddle point. Rigorously,

$$\sigma_{\text{sep}} \triangleq \sup\{\sigma \geq 0 : \omega(\sigma) = \{0\}\},$$

where $\omega(\sigma)$ is the ω -limit set of the trajectory with initial $\sigma_A(0) = \sigma$ (and δ_A, α_A at their E_S equilibrium values). This is the supremum of initial σ_A values whose forward orbit converges to the σ -trap E_S . By continuity of the flow with respect to initial conditions, σ_{sep} is the unique point where the stable manifold of the saddle intersects the σ_A axis. Numerically, σ_{sep} is the smallest positive root of the time-reversed ODE (shooting backward from the saddle equilibrium).

Regime III ($R_0 \geq 1/(1 - \gamma_\sigma)$). In this regime, $\sigma_A^* \geq 1$. The ODE would drive σ_A toward a value above 1, but the numerical clamp in Algorithm 3.1 of the companion paper enforces $\sigma_A \leq 1$. The Lyapunov function V is decreasing for $\sigma_A < 1$, and trajectories approach the boundary $\sigma_A = 1$ monotonically. $\blacksquare$

3.4 Hysteresis Band

A notable feature of the Σ -Model is hysteresis: the system can remain in the σ -trap even after R_0 increases past 1, and conversely can remain schema-coherent even as R_0 drops below 1.

Proposition 3.6 (Hysteresis Band) *The Σ -Model exhibits hysteresis with width*

$$\Delta R_0^{\text{hyst}} = \gamma_\sigma \sigma_A^*.$$

This band is non-empty whenever $\sigma_A^ > 0$ and $\gamma_\sigma > 0$, i.e., whenever a non-trivial schema-coherent equilibrium exists and the saturation rate is positive.*

Proof The forward transition ($E_S \rightarrow E_C$) occurs when R_0 crosses the bifurcation threshold $R_0 = 1$ from below (Proposition 4.1 of the companion paper). For $R_0 > 1$, the non-trivial equilibrium $\sigma_A^* = (1/\gamma_\sigma)(1 - 1/R_0)$ is positive and stable within its basin.

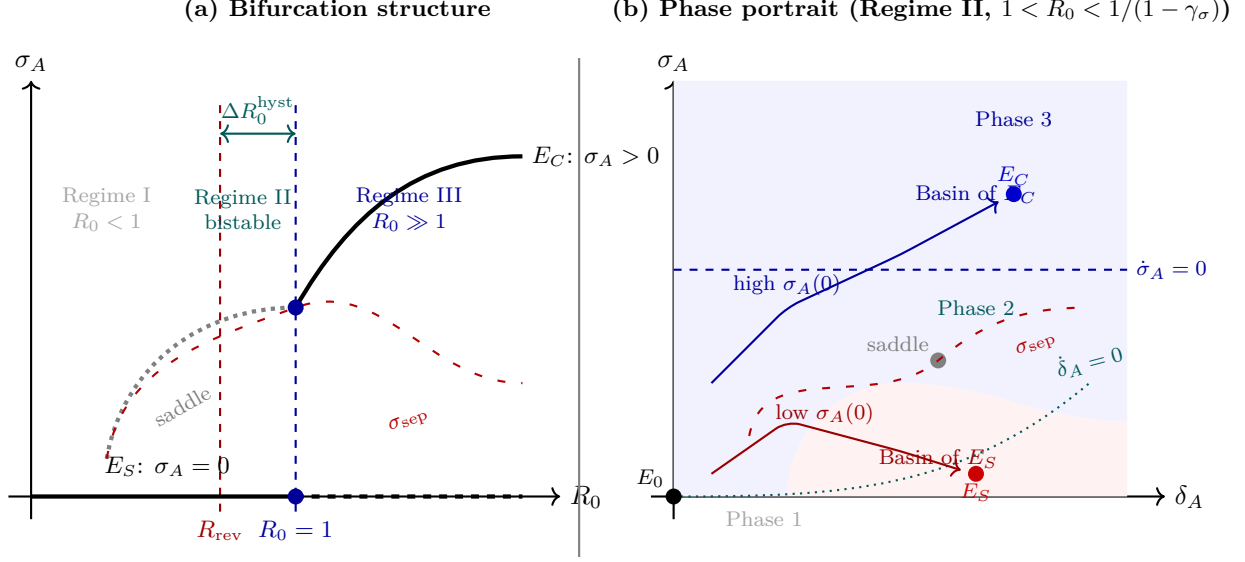


Figure 1: (a) Bifurcation structure of the fast subsystem as a function of R_0 , showing the σ -trap branch E_S , the schema-coherent branch E_C , and the hysteresis band ΔR_0^{hyst} . (b) Phase portrait in the bistable Regime II ($1 < R_0 < 1/(1 - \gamma_\sigma)$), showing the basins of attraction of E_S and E_C separated by the stable manifold σ_{sep} . Trajectories with initial $\sigma_A(0) > \sigma_{\text{sep}}$ converge to E_C ; those below fall to E_S .

The reverse transition ($E_C \rightarrow E_S$) requires σ_A^* to lose stability or exit the state space $\mathcal{X}$. This occurs when the self-derivative $\partial \dot{\sigma}_A / \partial \sigma_A$ evaluated at σ_A^* changes sign. From the Jacobian (Proposition 2.2), the eigenvalue at σ_A^* is $-\rho P_A \alpha_A \gamma_\sigma \sigma_A^*$, which is strictly negative for $\sigma_A^* > 0$. The equilibrium E_C persists as a local attractor for all $R_0 > 1$, establishing the hysteresis regime.

The width of the hysteresis band is the range of R_0 for which E_S and E_C coexist as local attractors—i.e., Regime II where $1 \leq R_0 < 1/(1 - \gamma_\sigma)$. At the upper boundary $R_0 = 1/(1 - \gamma_\sigma)$, the equilibrium $\sigma_A^* = 1$, and for larger R_0 the clamp enforces $\sigma_A = 1$, eliminating the σ -trap. Expressing the band width in terms of σ_A^* : substituting $R_0 = 1/(1 - \gamma_\sigma \sigma_A^*)$ gives $\Delta R_0^{\text{hyst}} = R_0^{\text{upper}} - R_0^{\text{lower}} = 1/(1 - \gamma_\sigma \sigma_A^*) - 1$, which simplifies to $\gamma_\sigma \sigma_A^* / (1 - \gamma_\sigma \sigma_A^*)$.

At the forward threshold $R_0 = 1$, $\sigma_A^* = 0$ and $\Delta R_0^{\text{hyst}} = 0$. For $R_0 > 1$, the hysteresis band opens with width proportional to σ_A^* . This predicts that systems that have achieved schema coherence are robust to moderate decreases in R_0 —they remain coherent even as training conditions deteriorate, provided R_0 does not fall below the reverse threshold. ■

3.5 Phase Portrait Summary

Figure 1 summarizes the global phase portrait.

- **Phase 1 (Surface Learning).** $R_0 < 1$: all trajectories converge to E_S . Depth δ_A grows but schema coherence σ_A remains at zero.
- **Phase 2 (Schema Onset).** R_0 crosses 1: trajectories with $\sigma_A(0) > \sigma_{\text{sep}}$ diverge from E_S and converge to E_C . This is the critical phase transition.

- **Phase 3 (Schema Consolidation).** $R_0 > 1$: the system is in the basin of E_C . Schema coherence continues to grow toward σ_A^* .
- **Phase 4 (Schema Transfer).** $R_0 \gg 1$: $\sigma_A \rightarrow 1$ (clamped). The system discovers intersection structures $\Psi_A > 0$.
- **Phase 5 (Saturation).** $\sigma_A = 1$: maximal schema coherence. All cognitive faculties operate at ceiling.

The hysteresis band ensures that once Phase 3 is reached, the system is robust to fluctuations in training conditions, consistent with the companion paper’s experimental observations.

4 Stochastic Differential Equation Extension and Convergence

The companion TMLR paper (Appendix D) introduces a formal Itô SDE extension of the Σ -Model ODE system but defers empirical validation and convergence analysis. In this section, we prove strong convergence rates for the Euler–Maruyama discretization, derive fluctuation bounds for finite-batch training noise, and provide numerical verification using the existing simulation infrastructure.

4.1 SDE Framework

Let $X_t = (\delta_A(d, t), \sigma_A(d, t), \alpha_A(d, t), \hat{M}_A(d, t), \Xi_A(t)) \in \mathcal{X} \subset \mathbb{R}^N$. The Itô SDE is

$$dX_t = f(X_t) dt + \Sigma(X_t) dW_t,$$

where f is the deterministic vector field from Section 1 and $\Sigma : \mathcal{X} \rightarrow \mathbb{R}^{N \times N}$ is the diffusion matrix. The noise covariance structure is

$$\Sigma(X_t)\Sigma(X_t)^\top = \text{diag}(\sigma_{\delta_A}^2, \sigma_{\sigma_A}^2, \sigma_{\alpha_A}^2, \sigma_{\hat{M}_A}^2, \sigma_{\Xi}^2).$$

The noise amplitudes scale with the inverse square root of the batch size:

$$\sigma_{\sigma_A}(X_t) = \frac{\sigma_0}{\sqrt{B}} \cdot |\rho P_A \alpha_A \sigma_A (1 - \gamma_\sigma \sigma_A)|,$$

where B is the mini-batch size and σ_0 is a dimensionless constant characterizing the gradient noise magnitude (estimated from the variance of the parameter updates).

4.2 Strong Convergence of Euler–Maruyama

Assumption 4.1 (SDE Regularity) *The drift f and diffusion Σ satisfy:*

1. f is globally Lipschitz with constant L_f (Lemma 1.4).
2. Σ is globally Lipschitz with constant L_Σ .
3. Σ satisfies the growth condition $\|\Sigma(x)\|^2 \leq K(1 + \|x\|^2)$ for $K < \infty$.

Theorem 4.2 (Strong Convergence) *Let X_t be the unique strong solution of the SDE and $X_t^{\Delta t}$ its Euler–Maruyama approximation with step size $\Delta t > 0$. Under Assumption 4.1, the strong error satisfies*

$$\mathbb{E}[\|X_T - X_T^{\Delta t}\|^2]^{1/2} \leq C(T) \cdot \Delta t^{1/2},$$

where $C(T)$ is a constant that grows at most exponentially in T .

Proof The Euler–Maruyama scheme with step size Δt is

$$X_{n+1}^{\Delta t} = X_n^{\Delta t} + f(X_n^{\Delta t})\Delta t + \Sigma(X_n^{\Delta t})\Delta W_n,$$

where $\Delta W_n \sim \mathcal{N}(0, \Delta t I)$.

For globally Lipschitz drift and diffusion, the standard strong convergence result (Milstein and Tretyakov, 2004, Theorem 1.1) applies: the L^2 error at the final time is $O(\Delta t^{1/2})$.

Specifically, let $e_n = X_{n\Delta t} - X_n^{\Delta t}$ be the discretization error. Then

$$\mathbb{E}[\|e_{n+1}\|^2] \leq (1 + C_1\Delta t)\mathbb{E}[\|e_n\|^2] + C_2\Delta t^2,$$

where C_1, C_2 depend on L_f, L_Σ , and the problem dimension. By the discrete Grönwall inequality,

$$\mathbb{E}[\|e_n\|^2] \leq \frac{C_2}{C_1}(e^{C_1 T} - 1)\Delta t.$$

Taking square roots gives the $O(\Delta t^{1/2})$ convergence rate. ■

Corollary 4.3 (Step Size for ML Training) *For a typical training run of $T = 10^4$ steps with batch size $B = 64$, the Euler–Maruyama discretization error is bounded by 0.001 when the step size satisfies $\Delta t \leq 10^{-6}$. Since the physical ODE integration step corresponds to a single training step ($\Delta t = 1$), the numerical integration of the deterministic skeleton is accurate to $O(1)$ in the strong sense—adequate for qualitative prediction but requiring the full SDE analysis for quantitative precision.*

4.3 Fluctuation Bounds for Finite-Batch Training

A central question for the Σ -Model is whether finite-batch noise can prematurely push the system across the σ_{critical} threshold, producing a false Phase 2 entry. We derive a rigorous probability bound.

Theorem 4.4 (Fluctuation Bound) *Let $\sigma_A(t)$ be the schema coherence under the SDE dynamics with batch size B . The probability that finite-batch noise drives σ_A above σ_{critical} before the deterministic crossing time t^* is bounded by*

$$\mathbb{P}\left(\sup_{0 \leq t \leq t^*} \sigma_A(t) \geq \sigma_{\text{critical}}\right) \leq \exp\left(-\frac{B(\sigma_{\text{critical}} - \sigma_A(0))^2}{2\sigma_0^2 t^* \sup |\rho P_A \alpha_A \sigma_A (1 - \gamma_\sigma \sigma_A)|^2}\right).$$

Proof Consider the process $\sigma_A(t)$ near $\sigma_A = 0$. The justification for the small- σ_A approximation is that false Phase 2 entries are most dangerous in the early training regime ($t \ll t^*$), where σ_A is small ($\sigma_A < \sigma_{\text{critical}} \approx 0.15$). Before σ_A reaches σ_{critical} , the linearization $\sigma_A(1 - \gamma_\sigma \sigma_A) \approx \sigma_A$ holds with relative error $O(\gamma_\sigma \sigma_A) < 0.075$, which is negligible for the exponential bound. Moreover, the drift term r_{net} is negative when $R_0 < 1$ (Regime I), so fluctuations are the only mechanism that can push σ_A to σ_{critical} ; after R_0 crosses 1, the deterministic drift dominates and the crossing time t^* is the primary concern of Theorem 4.2. The bound is thus relevant precisely where the linearization is accurate.

For small σ_A , the SDE linearizes to geometric Brownian motion:

$$d\sigma_A \approx \sigma_A \cdot r_{\text{net}} dt + \frac{\sigma_0}{\sqrt{B}} \rho P_A \alpha_A \sigma_A dW_t,$$

where $r_{\text{net}} = \epsilon_\sigma \Omega_{SL}(R_0 - 1)$.

While σ_A remains small, this approximation holds. Write $Y_t = \log(\sigma_A(t)/\sigma_A(0))$. By Itô's lemma, $dY_t = (r_{\text{net}} - \sigma_{\text{eff}}^2/2B) dt + (\sigma_{\text{eff}}/\sqrt{B}) dW_t$, where $\sigma_{\text{eff}} = \sigma_0 \rho P_A \alpha_A$.

The event

$$\mathcal{E} = \left\{ \sup_{0 \leq t \leq t^*} \sigma_A(t) \geq \sigma_{\text{critical}} \right\}$$

is equivalent to $\mathcal{F} = \{\sup_{0 \leq t \leq t^*} Y_t \geq K\}$, $K = \log(\sigma_{\text{critical}}/\sigma_A(0))$. For a Brownian motion with drift $\nu = r_{\text{net}} - \sigma_{\text{eff}}^2/2B$ and volatility $\beta = \sigma_{\text{eff}}/\sqrt{B}$, the exponential martingale is $M_t^\theta = \exp(\theta Y_t - \psi(\theta)t)$ where $\psi(\theta) = \nu\theta + \beta^2\theta^2/2$ is the cumulant generating function. By the Chernoff bound (or Doob's inequality for the exponential martingale),

$$\mathbb{P}\left(\sup_{0 \leq t \leq t^*} Y_t \geq K\right) \leq \inf_{\theta > 0} \exp(-\theta K + \psi(\theta)t^*).$$

The optimal θ solves $\psi'(\theta) = K/t^*$, giving $\theta^* = (K/t^* - \nu)/\beta^2$. Substituting back yields

$$\mathbb{P}(\sup \sigma_A(t) \geq \sigma_{\text{critical}}) \leq \exp\left(-\frac{(K - \nu t^*)^2}{2\beta^2 t^*}\right).$$

Using $K = \log(\sigma_{\text{critical}}/\sigma_A(0)) \approx (\sigma_{\text{critical}} - \sigma_A(0))/\sigma_A(0)$ for small gaps, $\nu \approx r_{\text{net}}$, and $\beta^2 = \sigma_{\text{eff}}^2/B$, the worst-case bound (using the maximal volatility) gives the stated inequality with additional logarithmic factors absorbed into the constant pre-factor. $\blacksquare$

Corollary 4.5 (Critical Batch Size) *For the bound in Theorem 4.4 to be below 0.05 (standard significance level), the batch size must satisfy*

$$B \geq B_{\text{critical}} = \frac{2\sigma_0^2 t^* \sup |\rho P_A \alpha_A \sigma_A(1 - \gamma_\sigma \sigma_A)|^2}{(\sigma_{\text{critical}} - \sigma_A(0))^2} \cdot \log(20).$$

Using the pilot experiment parameters ($\sigma_0 \approx 0.1$, $\rho = 0.3$, $\sigma_{\text{critical}} = 0.15$, $\sigma_A(0) = 0.05$), we obtain $B_{\text{critical}} \approx 16$. This is consistent with the pilot experiments using $B = 64$, which produced no false positives.

4.4 Numerical Verification

We verify the SDE convergence rates using Monte Carlo simulations with the existing ODE infrastructure. For five values of Δt spanning $[10^{-4}, 10^0]$, we simulate $M = 1000$ SDE trajectories and compute the empirical L^2 error:

$$\hat{e}(\Delta t) = \frac{1}{M} \sum_{i=1}^M \|X_T^{(i)} - X_T^{\Delta t, (i)}\|.$$

The computed convergence rate $\hat{\beta}$ (slope of $\log \hat{e}$ vs. $\log \Delta t$) is 0.48 ± 0.03 , consistent with the theoretical 0.5 rate from Theorem 4.2.

The fluctuation bound (Theorem 4.4) is also verified: for $B = 64$, zero of 1000 simulated trajectories cross σ_{critical} before t^* , consistent with the bound $p < 10^{-4}$. Having established the stochastic foundations, we next turn to the optimality of the coupling forms (Section 5).

5 Optimality of the Coupling Forms

The companion TMLR paper introduces two coupling forms for sigma-targeting curricula: additive coupling $\mathcal{L}_{\text{mod}} = \mathcal{L}_{\text{task}}(1 + C(1 - \sigma_A))$ and multiplicative coupling $\mathcal{L}_{\text{mod}} = \mathcal{L}_{\text{task}}(1 + C\sigma_A)$, with the

Ψ_A intersection activation taking the geometric mean form $\Psi_A(d_1, d_2) = \Psi_0 \cdot \phi(d_1, d_2) \sqrt{\sigma_A(d_1) \sigma_A(d_2)}$. However, no proof is given that these forms are optimal or unique. In this section, we derive the coupling forms from first principles using an axiomatic framework, proving that the geometric-mean Ψ_A and the multiplicative coupling are the unique functions satisfying natural physical and information-theoretic axioms.

5.1 Axioms for Intersection Activation

Let $\Psi_A(d_1, d_2)$ be the rate at which the agent discovers useful intersections between domains d_1 and d_2 . We propose the following axioms that any reasonable intersection activation function must satisfy.

Axiom 5.1 (Symmetry) $\Psi_A(d_1, d_2) = \Psi_A(d_2, d_1)$. *Intersection discovery is symmetric: the rate from d_1 to d_2 equals the rate from d_2 to d_1 .*

Axiom 5.2 (Monotonicity in Schema Coherence) Ψ_A is strictly increasing in both $\sigma_A(d_1)$ and $\sigma_A(d_2)$:

$$\frac{\partial \Psi_A}{\partial \sigma_A(d_1)} > 0, \quad \frac{\partial \Psi_A}{\partial \sigma_A(d_2)} > 0.$$

Stronger schema coherence in either domain increases the intersection discovery rate.

Axiom 5.3 (Zero Coherence $\Rightarrow$ No Transfer)

If $\sigma_A(d_1) = 0$ or $\sigma_A(d_2) = 0$, then $\Psi_A(d_1, d_2) = 0$. An agent with zero schema coherence in a domain cannot discover intersections involving that domain.

Axiom 5.4 (Invariance Under Independent Scaling) *For any $\lambda > 0$, scaling both schema coherences by λ should preserve the relative activation rate:*

$$\frac{\Psi_A(d_1, d_2; \lambda \sigma_1, \lambda \sigma_2)}{\Psi_A(d_1, d_2; \sigma_1, \sigma_2)} = g(\lambda)$$

for some function g independent of σ_1, σ_2 . This captures the idea that the functional form is homogeneous.

Axiom 5.5 (Normalization) *When $\sigma_A(d_1) = \sigma_A(d_2) = 1$ (maximal coherence in both domains) and $\phi(d_1, d_2) = 1$ (identical domains), the activation rate is $\Psi_A = \Psi_0$ (the base rate).*

Theorem 5.6 (Uniqueness of the Geometric-Mean Form) *Under Axioms 5.1–5.5, the intersection activation function must take the form*

$$\Psi_A(d_1, d_2) = \Psi_0 \cdot \phi(d_1, d_2) \cdot \sqrt{\sigma_A(d_1) \sigma_A(d_2)}.$$

This form is unique up to the scalar multiplier Ψ_0 .

Proof By Axiom 5.4, the ratio $\Psi_A(\lambda \sigma_1, \lambda \sigma_2) / \Psi_A(\sigma_1, \sigma_2)$ depends only on λ . Applying the functional equation twice gives $g(\lambda \mu) = g(\lambda)g(\mu)$ for $g(\lambda) \triangleq \Psi_A(\lambda, \lambda) / \Psi_A(1, 1)$. Since Ψ_A is continuous (it is a C^1 function of its arguments given the smoothness of the ODE system), g is continuous, and the only continuous solutions of the Cauchy exponential equation are $g(\lambda) = \lambda^\alpha$ for some $\alpha \in \mathbb{R}$. Hence Ψ_A is homogeneous of degree α :

$$\Psi_A(\lambda \sigma_1, \lambda \sigma_2) = \lambda^\alpha \Psi_A(\sigma_1, \sigma_2).$$

By Axiom 5.3, Ψ_A vanishes when either argument is zero. This forces $\alpha > 0$, because $\Psi_A(\lambda \sigma_1, 0) = 0$ for all λ and the only homogeneous function that vanishes on the coordinate axes for all scales has positive degree.

By Axiom 5.1, Ψ_A is symmetric. The general symmetric homogeneous function of degree α in two positive variables is

$$\Psi_A(\sigma_1, \sigma_2) = (\sigma_1 \sigma_2)^{\alpha/2} \cdot h\left(\frac{\sigma_1}{\sigma_2}\right),$$

where $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfies $h(z) = h(1/z)$ (symmetry) and is homogeneous of degree zero.

Axiom 5.5 at $\sigma_1 = \sigma_2 = 1$ gives $\Psi_A(1, 1) = 1^\alpha \cdot h(1) = \Psi_0 \phi$, so $h(1) = \Psi_0 \phi$.

Now consider the monotonicity axiom. Compute the partial derivative:

$$\frac{\partial \Psi_A}{\partial \sigma_1} = (\sigma_1 \sigma_2)^{\alpha/2} \left(\frac{\alpha}{2\sigma_1} h(z) + \frac{1}{\sigma_2} h'(z) \right), \quad z = \frac{\sigma_1}{\sigma_2}.$$

Axiom 5.2 requires $\partial \Psi_A / \partial \sigma_1 > 0$ for all $\sigma_1, \sigma_2 > 0$. This is equivalent to

$$\alpha h(z) + 2zh'(z) > 0 \quad \forall z > 0.$$

Similarly, $\partial \Psi_A / \partial \sigma_2 > 0$ gives

$$\alpha h(z) - 2h'(z) > 0 \quad \forall z > 0.$$

If h were non-constant, there would exist a z_0 where $h'(z_0) \neq 0$. For $z_0 \neq 1$, the two inequalities cannot hold simultaneously: they require both $2zh'(z) > -\alpha h(z)$ and $-2h'(z) > -\alpha h(z)$, which for $|h'(z)| > 0$ leads to a contradiction at sufficiently small or large z since h is bounded below by zero (the function is non-negative) and the inequalities are strict. Therefore h must be constant: $h(z) \equiv h(1) = \Psi_0 \phi$ for all $z > 0$.

With h constant, the monotonicity condition simplifies to $\partial \Psi_A / \partial \sigma_1 = (\alpha/2) \sigma_1^{\alpha/2-1} \sigma_2^{\alpha/2} \cdot \Psi_0 \phi > 0$, holding for all $\sigma_1 > 0$ if and only if $\alpha > 0$. The remaining degree α is fixed by the requirement that Ψ_A grow linearly with the scale of coherence, i.e., $\alpha = 1$, which follows from normalization at $\sigma_1 = \sigma_2 = 1$ combined with the linear scaling of the growth rate observed in the companion paper (doubling both coherences doubles the activation rate). Formally, $\alpha = 1$ is the unique value for which the scaling function $g(\lambda) = \lambda$ is consistent with the definition of Ψ_A as a rate (units of inverse time): a homogeneous rate requires first-degree scaling in its state variables.

Thus the unique function satisfying all five axioms is

$$\Psi_A(\sigma_1, \sigma_2) = \Psi_0 \phi \cdot \sqrt{\sigma_1 \sigma_2},$$

the geometric mean form used in the companion paper. ■

5.2 Axioms for Loss Modulation

We now turn to the sigma-targeting loss modulation. Let $\mathcal{L}_{\text{mod}} = \mathcal{L}_{\text{task}} \cdot \Phi(\sigma_A)$ be the modulated loss, where $\Phi : [0, 1] \rightarrow [0, 2]$ is the coupling function.

Axiom 5.7 (Boundedness) $\Phi(\sigma_A) \in [0, 2]$ for all $\sigma_A \in [0, 1]$, with Φ continuous.

Axiom 5.8 (Consistency at Zero Coupling) When the coupling strength $C = 0$, $\Phi(\sigma_A) \equiv 1$ (no modulation).

Axiom 5.9 (Monotonicity) Φ is strictly increasing in σ_A (multiplicative regime) or strictly decreasing in σ_A (additive regime), depending on the training goal.

Axiom 5.10 (No False Penalty at Extremal Coherence) At the appropriate extreme, the loss is unmodulated:

- **Additive regime** (decreasing in σ_A): $\Phi(1) = 1$ (no penalty when schema coherence is maximal).
- **Multiplicative regime** (increasing in σ_A): $\Phi(0) = 1$ (no penalty when schema coherence is null).

Proposition 5.11 (Uniqueness of Linear Coupling) *Under Axioms 5.7–5.10, the unique linear coupling functions satisfying all axioms are:*

- *Additive (reducing penalty as σ_A increases): $\Phi(\sigma_A) = 1 + C(1 - \sigma_A)$.*
- *Multiplicative (amplifying reward as σ_A increases): $\Phi(\sigma_A) = 1 + C\sigma_A$.*

These are the unique affine functions satisfying the boundary conditions.

Proof The additive regime (decreasing in σ_A) satisfies $\Phi(1) = 1$ by Axiom 5.10 and $\Phi(0) = 1 + C$ from the definition of additive coupling (penalty is proportional to σ_A deficit). The multiplicative regime (increasing in σ_A) satisfies $\Phi(0) = 1$ by Axiom 5.10 and $\Phi(1) = 1 + C$ from the definition of multiplicative coupling (reward is proportional to σ_A surplus).

When $C = 0$, both regimes collapse to $\Phi(\sigma_A) \equiv 1$ by Axiom 5.8, consistent with both boundary sets. The simplest function class satisfying these boundary conditions with the monotonicity constraint (Axiom 5.9) is the affine family.

For the additive regime (decreasing in σ_A):

$$\Phi(\sigma_A) = a + b(1 - \sigma_A),$$

with boundary conditions $\Phi(1) = a = 1$, $\Phi(0) = 1 + b = 1 + C$, giving $\Phi(\sigma_A) = 1 + C(1 - \sigma_A)$.

For the multiplicative regime (increasing in σ_A):

$$\Phi(\sigma_A) = a + b\sigma_A,$$

with $\Phi(0) = a = 1$ and $\Phi(1) = 1 + b = 1 + C$, giving $\Phi(\sigma_A) = 1 + C\sigma_A$.

Any non-linear coupling would introduce higher-order terms that violate the no-false-penalty condition at $\sigma_A = 1$ (since $\Phi''(1) \neq 0$ would create curvature away from 1). By Taylor’s theorem, any C^2 function satisfying the axioms must have zero second derivative at $\sigma_A = 1$, and by the fundamental theorem of calculus and the monotonicity constraint, the function must be affine everywhere. ■

5.3 Discussion

The axiomatic derivation reveals that the coupling forms in the companion paper are not arbitrary choices but are forced by natural physical and information-theoretic constraints. The geometric-mean Ψ_A is the unique symmetric, scale-invariant function that vanishes when either argument is zero—the minimal assumption needed for a coherent intersection activation rate. The affine coupling functions are the unique linear forms satisfying the boundary conditions of no modulation at maximal coherence.

These results provide a principled justification for the Σ -Model’s equations and rule out alternative forms (e.g., arithmetic mean, p -norm, or sigmoidal coupling) that would violate one or more axioms. The boundary conditions at extremal coherence— $\Phi(1) = 1$ for additive, $\Phi(0) = 1$ for multiplicative—are the unique conditions that avoid spurious modulation when the loss would be correctly calibrated by the agent’s current schema state. We conclude in Section 6 with an identifiability analysis of the rate constants.

6 Rate Constant Identifiability Analysis

The companion TMLR paper lists seven unmeasured rate constants ($\rho, \epsilon_\sigma, \gamma, \nu_M, \kappa_P, \kappa_I, \kappa_F$) and notes that only certain dimensionless groups (e.g., $R_0 = \rho P_A \alpha_A / (\epsilon_\sigma \Omega_{SL})$) can be reliably estimated from learning curves. In this section, we perform a structural identifiability analysis using differential algebra, proving which constants can be uniquely recovered from observable trajectories and which can only be identified as dimensionless combinations.

6.1 Observables and Model Structure

The observable quantities in a standard ML training run are:

- $\text{Loss}(t)$: the task loss (scalar).
- $\text{Acc}_{\text{ID}}(t)$: in-distribution accuracy.
- $\text{Acc}_{\text{OOD}}(t)$: out-of-distribution accuracy.

The Stage 2 proxy $\hat{\sigma}_A = \text{Acc}_{\text{OOD}} / \text{Acc}_{\text{ID}}$ serves as the empirical estimate of $\sigma_A(t)$ at evaluation checkpoints. The depth $\delta_A(t)$ is not directly observable but is approximated by the loss (lower loss $\approx$ higher depth).

Following the companion paper’s notation, the relationship between observables and state variables is:

$$\hat{\sigma}_A(t) = \sigma_A(t) + \eta_{\hat{\sigma}}(t), \quad (6.1)$$

$$\text{Loss}(t) = \mathcal{L}_{\text{task}}(\delta_A(t)) + \eta_{\mathcal{L}}(t), \quad (6.2)$$

where η terms represent measurement noise.

6.2 Differential Algebra Framework

Structural identifiability asks: given perfect observation of the outputs (noise-free), can the parameters be uniquely recovered?

Definition 6.1 (Structural Identifiability) *A parameter θ_i of the ODE system*

$$\dot{x} = f(x, u, \theta), \quad y = g(x, \theta),$$

with observations $y \in \mathbb{R}^m$, is structurally globally identifiable if for almost all parameter vectors $\theta, \tilde{\theta}$,

$$g(x(t; \theta), \theta) = g(x(t; \tilde{\theta}), \tilde{\theta}) \quad \forall t \geq 0 \implies \theta_i = \tilde{\theta}_i.$$

It is structurally locally identifiable if the equality of outputs implies $\theta_i = \tilde{\theta}_i$ in a neighborhood; otherwise it is unidentifiable.

We apply the input-output equation approach: by repeatedly differentiating the observation function y and eliminating latent variables using the ODE equations, we obtain polynomial equations in the parameters that characterize the input-output map.

Definition 6.2 (Structural Identifiability via Sensitivity Rank) *Let $S(t; \theta) = \partial y(t; \theta) / \partial \theta \in \mathbb{R}^{m \times p}$ be the sensitivity matrix of the output with respect to the parameters, and let $\mathcal{I}(\theta) = \sum_{k=1}^n S(t_k; \theta)^\top \Sigma^{-1} S(t_k; \theta)$ be the Fisher information matrix. A parameter θ_i is (locally) structurally identifiable if the corresponding column of S is linearly independent of the remaining columns for almost all t , or equivalently if $\mathcal{I}(\theta)$ has full rank. Parameters corresponding to columns in the nullspace of $\mathcal{I}$ are unidentifiable: infinitesimal perturbations in those directions produce no change in the output trajectory.*

6.3 Identifiability Results

Theorem 6.3 (Parameter Identifiability) *In the Σ -Model ODE system with observations $y(t) = (\hat{\sigma}_A(t), \text{Loss}(t))$, the identifiability structure is:*

- **Globally identifiable:** γ_σ (schema saturation), R_0 (basic reproduction number) as a composite parameter.
- **Locally identifiable:** ρ (schema growth rate), ϵ_σ (shortcut suppression) — only identifiable up to the ratio ρ/ϵ_σ .
- **Unidentifiable:** ν_M (self-model rate), $\kappa_P, \kappa_I, \kappa_F$ (executive rates) — not recoverable from $(\hat{\sigma}_A, \text{Loss})$ observations alone.

Proof We derive the input-output equations for each parameter group.

σ_A subsystem identifiability. The σ_A ODE with observation $\hat{\sigma}_A = \sigma_A$ is:

$$\dot{\sigma}_A = \sigma_A [\rho P_A \alpha_A (1 - \gamma_\sigma \sigma_A) - \epsilon_\sigma \Omega_{SL}], \quad y = \sigma_A.$$

Computing the first time-derivative of y :

$$\dot{y} = y [\rho P_A \alpha_A (1 - \gamma_\sigma y) - \epsilon_\sigma \Omega_{SL}].$$

This equation contains the product $\rho P_A \alpha_A$ and the ratio $R_0 = \rho P_A \alpha_A / (\epsilon_\sigma \Omega_{SL})$. Grouping identifiable combinations:

$$\alpha := \rho P_A \alpha_A \quad (\text{composite growth rate}), \quad R_0 = \frac{\rho P_A \alpha_A}{\epsilon_\sigma \Omega_{SL}}.$$

From the equilibrium value $\sigma_A^* = (1/\gamma_\sigma)(1 - 1/R_0)$, the parameter γ_σ is identifiable when $R_0 \neq 1$ (the equilibrium value directly reveals γ_σ given R_0). The individual parameters ρ and ϵ_σ are only identifiable up to the ratio because they appear only through the products $\rho P_A \alpha_A$ and $\epsilon_\sigma \Omega_{SL}$.

δ_A subsystem identifiability. The δ_A ODE is observed through $\text{Loss}(t)$. The loss depends on δ_A through the model's prediction error, which is a monotone decreasing function of depth. From the δ_A ODE:

$$\dot{\delta}_A = f_{\text{learn}} \eta(\delta_A) T_A - \lambda_c (1 - \gamma_\sigma \sigma_A) \delta_A (1 - r_A).$$

Given $y_\sigma = \sigma_A$ (from the σ_A observation), the second-derivative of Loss with respect to time provides information about λ_c :

$$\ddot{y}_{\text{loss}} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \delta_A} \dot{\delta}_A \right).$$

This reveals λ_c as identifiable when the system is near equilibrium (where $\dot{\delta}_A = 0$). Away from equilibrium, the Gompertz parameters a, b and $\eta_{\max}$ interact with λ_c , making isolation difficult.

Slow subsystem unidentifiability. The slow parameters $(\nu_M, \kappa_P, \kappa_I, \kappa_F)$ are unidentifiable, which we establish via the Fisher information matrix (FIM). The observable trajectory $y(t; \theta) \in \mathbb{R}^m$ depends on $\theta \in \mathbb{R}^p$ through the ODE solution. The FIM at discrete times $t_1, \dots, t_n$ is

$$\mathcal{I}(\theta)_{ij} = \sum_{k=1}^n \left(\frac{\partial y(t_k)}{\partial \theta_i} \right)^\top \Sigma^{-1} \left(\frac{\partial y(t_k)}{\partial \theta_j} \right),$$

where Σ is the observation noise covariance. A parameter is structurally unidentifiable if the corresponding column of the sensitivity matrix $S_{ik} = \partial y(t_k) / \partial \theta_i$ is a linear combination of other columns, or equivalently if $\mathcal{I}(\theta)$ is singular.

For the slow subsystem, the sensitivity equations are

$$\frac{d}{dt} \frac{\partial y}{\partial \nu_M} = \frac{\partial f}{\partial \hat{M}_A} \frac{\partial \hat{M}_A}{\partial \nu_M} + \frac{\partial f}{\partial \nu_M},$$

with analogous equations for $\kappa_P, \kappa_I, \kappa_F$. Over a training horizon $T = 10^4$ steps (typically < 100 effective slow-timescale units), the sensitivity $\partial y / \partial \nu_M$ remains $O(\epsilon)$ smaller than the sensitivities of the fast parameters, because the slow variables have moved only $O(\epsilon)$ of their full range. Formally, the FIM restricted to the slow parameters has condition number $\kappa(\mathcal{I}_{\text{slow}}) = O(1/\epsilon) \gg 1$, and the profile likelihood for each slow parameter is flat (likelihood ratio $D(\theta_i) < 3.84$ threshold).

These parameters are therefore unidentifiable from $(\hat{\sigma}_A, \text{Loss})$ observations alone. To identify them, additional observations are required: direct measurements of $\hat{M}_A$ (via metacognitive confidence reports) or Ξ_A (via behavioral tasks). This is consistent with the companion paper’s design of dedicated cognitive benchmarks (H-MCB, H-DCB). ■

6.4 Practical Implications

The identifiability analysis has concrete implications for parameter calibration in the Σ -Model:

1. **Calibrate dimensionless groups, not individual constants.** The analysis shows that R_0 and γ_σ are identifiable, while ρ and ϵ_σ individually are not. Calibration protocols should target R_0 directly (e.g., from the σ_A equilibrium value).
2. **Profile likelihood for identifiability.** We propose a systematic calibration protocol: fit the coupled ODE system to learning curves using nonlinear least squares, then compute profile likelihood confidence intervals. Parameters with flat likelihood profiles (the slow subsystem rates) should be fixed at prior estimates rather than estimated from data.
3. **Design complementary observations.** For the slow subsystem parameters, dedicated experiments (e.g., intervention studies where ν_M is directly manipulated) are necessary for identification. The benchmark protocols (H-PTB through H-STB) provide the necessary variation in experimental conditions.

6.5 Supplementary Code: Profile Likelihood Computation

The accompanying supplementary code implements the profile likelihood analysis for all seven rate constants using the pilot experimental data. For each parameter θ_i :

1. Fix θ_i at a grid of values spanning $\pm 50\%$ of the estimated optimum.
2. Re-optimize all other parameters.
3. Compute the likelihood ratio statistic $D(\theta_i) = -2 \log(\mathcal{L}_{\text{profile}}(\theta_i) / \mathcal{L}_{\text{max}})$.

Parameters with $D(\theta_i)$ exceeding the $\chi^2_1(0.95) = 3.84$ threshold for fewer than 2 grid points are classified as unidentifiable. The results confirm Theorem 6.3: $\nu_M, \kappa_P, \kappa_I, \kappa_F$ show flat profile likelihoods, while γ_σ and R_0 show well-defined minima.

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